

Preclass Learning: Review, Setup, and Git (ICA 2)

1. Review of Foundational R/Tidyverse

- **Data Structure:** Basic understanding of data frames.
- **dplyr Essentials:** Fluency in the "five verbs" for basic data manipulation:
 - `select()`: Choosing columns.
 - `filter()`: Choosing rows based on conditions.
 - `mutate()`: Creating new columns.
 - `arrange()`: Sorting rows.
 - `summarize()`: Aggregating data (usually paired with `group_by()`).

2. Project-Oriented Workflow and File Organization

- **Project Root:** All file paths must be defined relative to the project's root directory (e.g., using `here::here()` in R). This ensures code works on any machine.
- **Standard Subdirectories:** Organizing files into dedicated, consistent folders:
 - `data/`: Raw, immutable data.
 - `scripts/`: All R/Python source code and functions.
 - `reports/`: Final outputs, presentations, and reports.

3. Git and GitHub for Version Control

- **Git:** A local version control system that tracks changes to files over time.
- **Repository (Repo):** The project folder tracked by Git.
- **Core Workflow:**
 - **Commit:** Taking a labeled snapshot of the project's current state.
 - **Push:** Uploading committed changes from the local repository to the remote repository (GitHub).
 - **Pull:** Downloading and integrating changes from the remote repository to the local machine.
 - **Branching:** Creating parallel development lines to isolate feature work from the main, stable code.



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